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NISAR DATA NAMING CONVENTION:-

Detailed Filename Breakdown:- Standard Science products (L1-L2) uses Multifield Convention.

→ NISAR_L_RSLC_001_015_A_025_1250_SH_A_20251020T120000_20251020T120030_D00101_P_F_J_001.h5

Mission Identifier:- NISAR

Frequency Band:- L or S Band

Product Type:- RSLC, GCOV,

Cycle & Relative orbit:- 001_015 (repeat Cycle: 001
Relative orbit: 015)

Pass Direction:- A (ascending), D (Descending)

Frame Number:- 025 (Unique Identifier for data 'Slice')

Radar Mode:- 1250 (Centre frequency & Bandwidth)

Polarization:- SH, DH, QP (Quad Pol), QQP (Quasi Quad Pol)

Start & End Time:- UTC (YYYYMMDDTHHMMSS)

Composite Release ID:- (CRID):- D00101
(Software & Calibration Versions used).

Internal File Structure (HDF5):-
NISAR, .h5 are hierarchical Containers.

Science Group:- Contain primary raster data ("Back Scatter")

b) Metadata Group:- Stores Sensor parameters orbit State Vectors & processing history, orbit Ephemeris.

c) Grid Group :- Provide high resolution Geolocated Coordinate Information. (lat/lon).

d) Quality Group:- Includes Noise Equivalent Sigma zero (NESZ), layout Masks & Local Incidence angle Layers., Quality Metrics (QA data) to help users evaluate data reliability.

3. Interferometric Product Variations:- Products from pairs (GUNW or RIFG) Modify Timing fields

- Ref Start/End Date Time:- Timestamps for first (reference) scene
- Sec Start/End Date Time:- Timestamps for Second (secondary/Match) Scene.

4. Simulated Data Naming (UAVSAR-based):- It is used for Testing following different legacy Inspired format.

Sitename_LineID Flight ID_DI_~~[illegible]~~ Band SquintPol_xtalkDither_NMode_Version.Ext.

a) Sitename:- 6 character Target Site abbreviation

b) LineID:- 5 Digit Track Identifier (header + Counter)

c) DT (Data Take):- Sequential Counter for aquisition & polarization Indicators

d) Bandsquint Pol:- Frequency band, Steering angle & Polarization Indicator.

NISAR DATA

NISAR (NASA-ISRO) SAR, data products are classified into several processing levels ranging from raw signal data to highlevel Geocoded Science products.

(A) LEVEL 0: Raw Data:

(1) • RRSD (Radar Raw data Signal):- Primary level 0B product corrected and aligned raw radar pulse data derived from Telemetry.

(2) LOA:- Intermediate time ordered raw data product used internally and not typically available for general dissemination.

(B) LEVEL 1:- (Focused range Doppler product):-

- RSLC: (Single Look Complex)- Focused SAR Imagery having phase and amplitude information. It is base product for all higher level science processing.

- RIFG: (Wrapped Interferogram):- Multilooked phase wrapped interferogram showing phase difference between two RSLC.

- RUNW:- (unwrapped Interferogram):- Phase unwrapped version of RIFG to calculate surface displacement.

- **ROFF:-** (Pixel offsets) :- Sub Pixels offset shifts between pair of RSLC's, used primarily for Measuring rapid Ground Motion.

(C) LEVEL 2:- Geocoded Products:-

- **GSLC:-** (Geocoded Single Look Complex) :- Map projected RSLC Imagery directly overlaid on Maps.

- **GUNW:-** (Geocoded unwrapped Interferogram) :- Analysis ready - displacement Maps resampled to Geocoded Grid.

- **GCOV:-** (Geocoded polarimetric Covariance) :- Multilooked ~~Radar~~ Backscatter data corrected for radiometric & Terrain Distortions for Surface characterization.

- **GOFF:-** (Geocoded Pixel offsets) :- Map projected version of pixel offsets, generated specifically for Cryosphere applications like Glacier flow Tracking.

LEVEL 3 and 4:- (Higher Level Science Products)

SM:- (Soil Moisture): Level 3 Global Soil Moisture product provided on a standard EASE Grid -2.

Science Maps:- Level 4 products Including Forest Biomass, inundation (wetland Mapping), Crop area Monitoring.

ACCESS & BETA DATA:-

Beta product:- Early release NISAR Beta Sample Data available via NASA Earth Data Search & ISRO's Bhoonidhi portal.

UR (Urgent Response):-(UR): In event of Disasters like Earthquakes or floods priority processed L0-L2 products are released through dedicated UR collections.

Applications (USE CASE):-

	PRODUCT	USE CASE	COORDINATES
1)	RRSD (L0)	Advanced process to reprocess raw signal data from Scratch	Raw Pulse.
2)	RSLC (L1)	InSAR, Measure mm level Ground deformation or building Subsidence.	RADAR
3)	RIFG/RUNW (L1)	Change detection, visualize phase shifts caused by Earthquakes or Volcanic activity.	RADAR
4)	GSLC (L2)	High precision Mapping – Complex data aligned to Map Coordinates.	
5)	GCOV (L2)	Land Cover & Ecosystems Monitor forest, agriculture, flood extents	Map(UTM)
6)	GUNW (L2)	Analysis ready deformation	Map(UTM)
7)	GOFF (L2)	Ice & Glaciers, Track fast Moving feature like Glacial flow	Map(UTM)
8)	SM(L3)	Agriculture/hydrology → Monitor Soil health	Ease-Grid.